The galaxy-bias model of the lssty1xdesi paper

One model and one prior for DESI C_ℓ^{gg} in thin redshift bins

November 2026

1 Summary

The question. What does DESI galaxy clustering, measured as angular power spectra C_ℓ^{gg} in thin redshift bins, constrain on its own, with a bias model a referee accepts? Thin bins resolve the growth of structure along the line of sight only if the bias of neighboring bins is not free bin by bin; tying the bins too tightly instead turns assumptions about the bias into cosmology.

The model. The fiducial model's galaxy power spectrum is one-loop – it keeps the first nonlinear corrections of perturbation theory – with seven bias quantities per bin: the linear, quadratic and tidal biases, the counterterms α_0 and α_2, and the stochastic terms SN_0 and SN_2. Each of them is its group's value, which follows calibrations where they exist, plus, depending on the quantity, a free tilt in redshift across the group and a residual of the bin with a Gaussian prior correlated in redshift (Section 5). The line-of-sight counterterm is fixed, $\alpha_2 = 0$ in every bin: the sampled Fingers-of-God dispersions – the smearing of each bin in redshift by the galaxies' random velocities – whose priors are widened six-fold, carry that freedom instead. One principle sets the priors: slicing a wide DESI sample into thin bins adds measurements but no knowledge of the bias, so the group value and the tilt carry exactly the freedom of the wide sample, and the residual is what the slicing may add within it (Section 6).

The result. On the two thin layouts – $\Delta d_{\mathrm{com}} = 62$ Mpc and $\Delta z = 0.02$, the finest slicings of Section 3 – the figure of merit $\mathrm{FoM}(w_0, w_a)$ of DESI C_ℓ^{gg} alone, the inverse area of the (w_0, w_a) error ellipse, grows from 0.564 and 0.352 at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ to 6.70 and 7.95 at $0.2\,h\,\mathrm{Mpc}^{-1}$, below the 17.3 of LSST Y1 3x2pt alone; $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ stays below LSST Y1 as well (Section 8). Over random bias histories, the known drifts plus scatter drawn from the prior, the quoted error bars are honest at every cut and on every layout (Section 8).

Pairs beyond the list. Cross spectra of bins further apart than those of the pair list – the set of bin pairs whose cross spectra the data vector holds – would raise $\mathrm{FoM}(w_0, w_a)$ by up to a factor of 1.55 on the thin layouts, widening their lead over the wide $\Delta z = 0.2$ layout from 22–26 times its figure of merit to 28–37 at $k_{\max} = 0.2\,h\,\mathrm{Mpc}^{-1}$ (Section 4). The gain rests on the freedom the prior grants: each group's free bias level and slope mimic the growth history in the auto spectra, the added cross spectra separate the two, and with that level and slope held fixed most of the gain goes. The analysis keeps the current data vector; the longer lists are a comparison against it. What stays open is a finite prior on the levels and slopes.

What the grouping is worth. A per-bin prior that looks looser than the model gains information by accident, from the slicing alone, up to a factor of 1.54 in $\mathrm{FoM}(w_0, w_a)$. Untied bins that keep the freedom of the wide samples cost a factor of 1.12 to 1.17 in $\mathrm{FoM}(w_0, w_a)$ on the thin layouts at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ with independent per-bin terms, and up to 1.8 with residuals correlated in redshift (Section 9). One constant value of every bias quantity per wide sample claims up to 7.5 times the model's $\mathrm{FoM}(w_0, w_a)$, with error bars up to 6.4 times too small.

2 Fiducial model and parameter values

Every number of this note was computed at the fiducial below, the evaluate point – the parameter values at which the data vectors and derivatives are evaluated – of the paper configurations `EXAMPLE_EVALUATE1–4.yaml`, frozen here so the note does not depend on files that may change.

Cosmology. Flat w_0w_aCDM: $10^9A_s = 2.1$, $n_s = 0.96605$, $H_0 = 67.32$ km/s/Mpc, $\Omega_b = 0.04$, $\Omega_m = 0.3$ (neutrinos included), $w_0 = -1$ and $w_0 + w_a = -1$, so $w_a = 0$; derived $\sigma_8 = 0.8278$. The Fisher matrices vary 10^9A_s, n_s, H_0, Ω_b, Ω_m, w_0 and w_a with flat priors; the neutrino mass sum stays at 0.06 eV and the optical depth at $\tau = 0.0697$.

Layouts. Every layout carries 53 band powers per spectrum over $15 \leq \ell \leq 544$ (Section 3).

layout	lens bins	cross pairs	entries
$\Delta d_{\rm com} = 62$ Mpc	76	107	10,494
$\Delta z = 0.02$	90	133	12,614
$\Delta z = 0.05$	36	52	5,459
$\Delta z = 0.2$	10	13	2,014

The entries count the stored data vector: 53 band powers for each auto and cross spectrum, plus 795 entries the data format reserves for cosmic-shear blocks and keeps at zero for DESI C_ℓ^{gg} alone – on $\Delta d_{\rm com} = 62$ Mpc, $53\,(76 + 107) + 795 = 10{,}494$ (`notes/fisher_machinery`).

Bias. The group amplitudes are $B_g = 1.34$ (BGS), 1.7 (LRG1, LRG2, LRG3) and 0.84 (ELG1, ELG2), each flat in $[0.8, 4]$; at the fiducial every tilt and residual is zero, so bin i has $b_{1,i} = B_g/D(\bar{z}_i)$, with the growth factor D read off the linear power spectrum at $k = 0.005$ Mpc^{-1}. The quadratic and tidal biases sit on their calibrations, Eq. (6) with zero offsets, and every counterterm and stochastic amplitude is zero: $\alpha_0 = \alpha_2 = \mathrm{SN}_0 = \mathrm{SN}_2 = 0$. The linear-bias residual prior has $\sigma_g = 3\%$ in every group and $L = 0.05$ (Section 6).

Fingers-of-God dispersions. Seven dispersion groups: the six bias groups, with BGS split at $\bar{z} = 0.255$ into BGS0 and BGS1. Each has a Gaussian prior centered on its fiducial, with a width six times its base width (Section 5):

[km/s]	BGS0	BGS1	LRG1	LRG2	LRG3	ELG1	ELG2
fiducial (prior center)	371	358	357	348	343	304	289
base width	145	105	70	50	50	40	30
prior width	870	630	420	300	300	240	180

The sampled priors are truncated at zero. The truncation matters most for BGS0, whose prior width exceeds its center: the truncated prior's mean is about 847 km/s, not the 371 km/s center. The Fisher forecast treats every σ_v prior as an untruncated Gaussian of the stated width.

Scale cuts. The UV-fraction cuts of every figure (Section 3) are computed on this fiducial, from the paper configurations, at the final derivative tier of Section 7 – the settings of the derivative base – and cached in

`chains/model_scale_cuts_final.npz`.

Covariance. Gaussian, computed at this same fiducial with massless neutrinos (the only difference from the data vector), at the fiducial dispersions, for 14,000 square degrees; every Fisher matrix rescales its inverse to 12,300 (Section 7).

Derivative tier. Five-point stencils at relative step $h = 0.02$, at the final settings of the convergence study (Section 7): grid factor $g = 256$, CosmoLike `accuracyboost` 3 and `integration_accuracy` 1, CAMB `AccuracyBoost` 2.0, `k_per_logint` 30, maximum wavenumber 60 Mpc^{-1} and the halofit tolerance patch. The base is cached in

`chains/model_derivatives_<layout>_final.npz`
and `chains/model_templates_<layout>_final.npz`,

one pair per layout above.

The LSST Y1 3x2pt reference. The reference values 17.3 and 215 are the lsst_y1 project's real-space Fisher forecast of the SRD 3x2pt analysis. Its fiducial differs from this note's: $w_0 = -0.9$, and nonzero photo-z-shift and shear-calibration centers, under Gaussian priors of widths 0.002 and 0.005.

3 Data and scale cut

The data are the DESI angular galaxy power spectra C_ℓ^{gg} in 53 bands of width $\Delta\ell = 10$ over $15 \le \ell \le 544$, with ℓ the multipole (a band power is one spectrum averaged over one band): the auto spectrum of every redshift bin and the cross spectra of neighboring and overlapping bins. On $\Delta d_{\mathrm{com}} = 62$ Mpc these are 107 pairs: the 97 of the DESI forecast's pair list and ten LRG3×ELG1 pairs at the same redshift, whose Fingers-of-God-blurred kernels overlap (Section 4). The galaxies form six tracer groups g, each a wide sample selected as a whole: BGS (the Bright Galaxy Survey), LRG1–3 (luminous red galaxies) and ELG1–2 (emission-line galaxies). Four layouts slice them: $\Delta d_{\mathrm{com}} = 62$ Mpc, 76 bins of equal comoving width, and $\Delta z = 0.02$, 0.05 and 0.2, with 90, 36 and 10 bins of fixed redshift width. The first two are the thin layouts: a bin of comoving width $\Delta\chi$ resolves line-of-sight wavenumbers up to $k_\parallel \simeq \pi/\Delta\chi$, about $0.075\,h\,\mathrm{Mpc}^{-1}$ for the 62 Mpc bins ($\pi/62\,\mathrm{Mpc} = 0.051\,\mathrm{Mpc}^{-1}$, or $0.051/h = 0.075\,h\,\mathrm{Mpc}^{-1}$), so thin bins keep much of the three-dimensional information [1]. Bin i has mean redshift $\bar{z}_i$, and $D(z)$ is the linear growth factor, $D(0) = 1$.

Parameters and figures of merit. Seven cosmological parameters vary with flat priors: $10^9 A_s$, with A_s the amplitude of the primordial curvature power spectrum; the spectral index n_s; the Hubble constant H_0; the baryon and matter densities today, Ω_b and Ω_m, in units of the critical density; and w_0 and w_a of the dark-energy equation of state $w_0 + w_a(1-a)$, with a the scale factor. Seven Fingers-of-God velocity dispersions σ_v vary under the Gaussian priors of Section 2, widened to carry the line-of-sight counterterm's freedom (Section 5): the line-of-sight peculiar velocities of galaxies – the bulk motion of their halos plus their motion inside them – shift the measured redshifts, and σ_v, the rms of those velocities, sets the width of the smearing this adds to a bin's redshift distribution (`notes/velocity_dispersion_per_bin`). With σ a parameter's marginalized one-sigma error – its error after every other parameter is fitted and its uncertainty folded in – $\mathrm{FoM}(w_0, w_a) = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$ is the inverse area of the marginalized (w_0, w_a) ellipse, and $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ is defined the same way. The reference is LSST Y1 3x2pt alone [3]: the lsst_y1

project's real-space Fisher forecast of the SRD analysis, whose fiducial differs from this note's (Section 2).

The scale cut. Matter is less nonlinear at higher redshift, so bin i is cut at $k_{\max}/D(\bar{z}_i)$, and every $k_{\max}$ here is the $z = 0$ value. The cut is the non-Limber UV-fraction criterion of Saraivanov et al. [4], after DeRose and Chen [5], evaluated with the exact (non-Limber) projection of Section 7: each band power splits into $C_\ell^{\rm IR}$, from the power spectrum below the bin's cut wavenumber, and $C_\ell^{\rm UV}$, from the power above it, and a band is kept while

$$\frac{|C_\ell^{\rm UV}|}{|C_\ell^{\rm UV}| + |C_\ell^{\rm IR}|} \;<\; 0.05 \tag{1}$$

and within the shipped mask, the k mask built into the data files before any scale cut, which stops every bin at $0.176\,h\,\mathrm{Mpc}^{-1}$. The cut is computed on the model's own fiducial (Section 2), at the final derivative tier, the settings of the derivative base. Computing it on linear-bias spectra instead moves the figures of merit by at most a few percent on the main layouts (`notes/scale_cut_fiducial_dependence`). The forecasts scan $k_{\max} = 0.05, 0.075, \ldots, 0.2\,h\,\mathrm{Mpc}^{-1}$.

4 The cross pairs and their overlaps

Which pairs enter the data vector. How correlated two bins' maps are is read off the overlap of their Fingers-of-God-convolved redshift distributions: with n_i the normalized distribution of bin i,

$$O_{ij} \;=\; \frac{\int dz\, n_i(z)\, n_j(z)}{\sqrt{\int dz\, n_i^2(z)\, \int dz\, n_j^2(z)}}, \tag{2}$$

1 for two identical shapes and 0 for disjoint ones. In the Limber regime – at high ℓ, where each band power reads the power spectrum only at one transverse wavenumber (Section 7) – the power spectrum and the distance barely change across two thin bins, so O_{ij} is close to the correlation coefficient $C_\ell^{ij}/\sqrt{C_\ell^{ii}\,C_\ell^{jj}}$ of the two spectra: it reads directly as how correlated the two maps are. The data vector holds DESI's measured pairs plus every pair with $O_{ij} \geq 10^{-6}$ at the fiducial dispersions, which adds the ten LRG3×ELG1 pairs of $\Delta d_{\rm com} = 62$ Mpc whose raw redshift ranges – the ranges before the Fingers-of-God blur – just miss. No unmeasured pair overlaps between 10^{-6} and 0.028, so the list does not depend on where in that window the threshold sits (`notes/cross_correlations`). This list is the current data vector, the reference of every ratio of this section.

What an overlap floor shows. Keeping only the pairs with $O_{ij} \geq O_{\min}$, measured ones included, drops for every floor up to 10^{-3} exactly one pair – the BGS×LRG1 boundary pair of $\Delta d_{\rm com} = 62$ Mpc, whose kernels overlap by 3.4×10^{-7} – at a cost of 0.2% of FoM(w_0, w_a) at $k_{\max} = 0.2\,h\,\mathrm{Mpc}^{-1}$ and nothing elsewhere (Figure 1). The list holds no pair between 3.4×10^{-7} and 3.4×10^{-3}, so the test does not fix where between those overlaps a pair begins to matter. A small overlap does not mean little information: at $O_{\min} = 10^{-2}$ only $\Delta z = 0.2$ loses pairs, whose overlaps are all below 10^{-2}, and its figures of merit fall to 0.89–0.95 of the current data vector's for (w_0, w_a) and 0.77–0.87 for $(\Omega_m, 10^9 A_s)$.

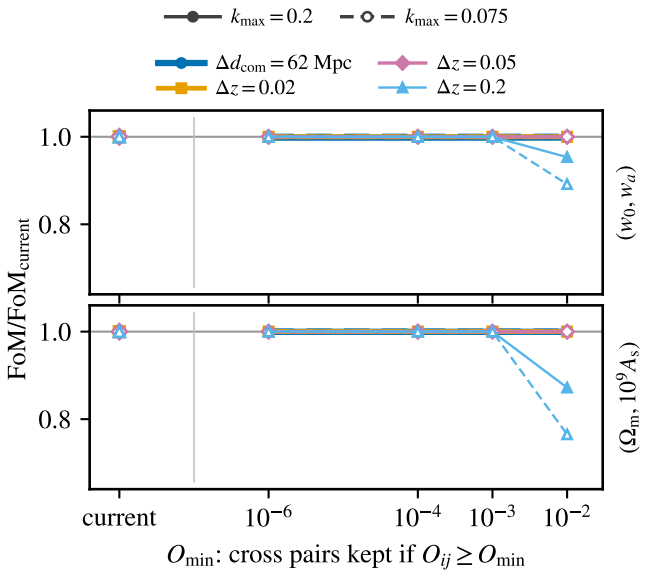


Figure 1. The figures of merit with only the cross pairs whose overlap at the fiducial dispersions reaches $O_{\rm min}$, over the current data vector's, against $O_{\rm min}$; "current", the current data vector, sits left of the separator. Upper panel: FoM(w_0, w_a); lower: FoM$(\Omega_m, 10^9 A_s)$; the panels share the y axis. Color, line width and marker: the layout; solid, filled markers: $k_{\max} = 0.2\,h\,{\rm Mpc}^{-1}$; dashed, open: 0.075; gray: one. $k_{\max}$ at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

Pairs beyond the list. The pair list and the covariance are built once, at the fiducial dispersions, while the model's six-times-wider priors allow fits in which the kernels blur further and unmeasured pairs would pass the overlap threshold. Two families of longer lists measure what such pairs are worth (Figure 2); each contains the current data vector and grows along one axis. The gap axis adds every pair of bins whose raw redshift ranges are at most g bin widths apart: each step in g adds the next chain of pairs, the spectra of bins two apart at $g = 1$, three apart at $g = 2$, and so on. The σ_v axis reruns the overlap criterion of Eq. (2) with every dispersion set to one common value s; s only selects the pairs, while the model and the covariance keep the fiducial dispersions, so this axis prices exactly the lists that a fit carrying the dispersions toward the upper edge of their priors would require. Those lists change late: on $\Delta d_{\rm com} = 62$ Mpc no pair joins until $s = 750$ km/s, on $\Delta z = 0.05$ none until 1200, and on $\Delta z = 0.2$ none ever; only $\Delta z = 0.02$ grows already at $s = 450$. On the three thinner layouts the added pairs are worth much: at its largest list, at the better of the two cuts the figure draws, each layout gains a factor of 1.55, 1.44 and 1.59 of FoM(w_0, w_a), in layout order, and up to 1.44 of FoM$(\Omega_m, 10^9 A_s)$, with most of the gain in the first added chain; on $\Delta z = 0.2$ the gain never exceeds 2%. (At $k_{\max} = 0.075\,h\,{\rm Mpc}^{-1}$ the scale cut keeps almost no band of the pairs added on $\Delta z = 0.02$, so that layout gains at the larger cuts.)

Why the added pairs help. Besides the seven cosmological parameters, the fit determines the nuisance parameters: the parameters that describe the galaxies rather than the universe – the biases, counterterms, stochastic amplitudes and dispersions – whose uncertainty the quoted errors include. The decisive ones here form each group's bias line: under the prior of Section 6 the bias amplitudes $\beta_i = b_{1,i} D(\bar{z}_i)$ of a group's bins follow a straight line in redshift with a free level and a free slope – the freedom of the wide sample – plus the correlated residuals the prior charges. The auto spectra alone cannot pin that line down. A higher level raises every auto spectrum of the group

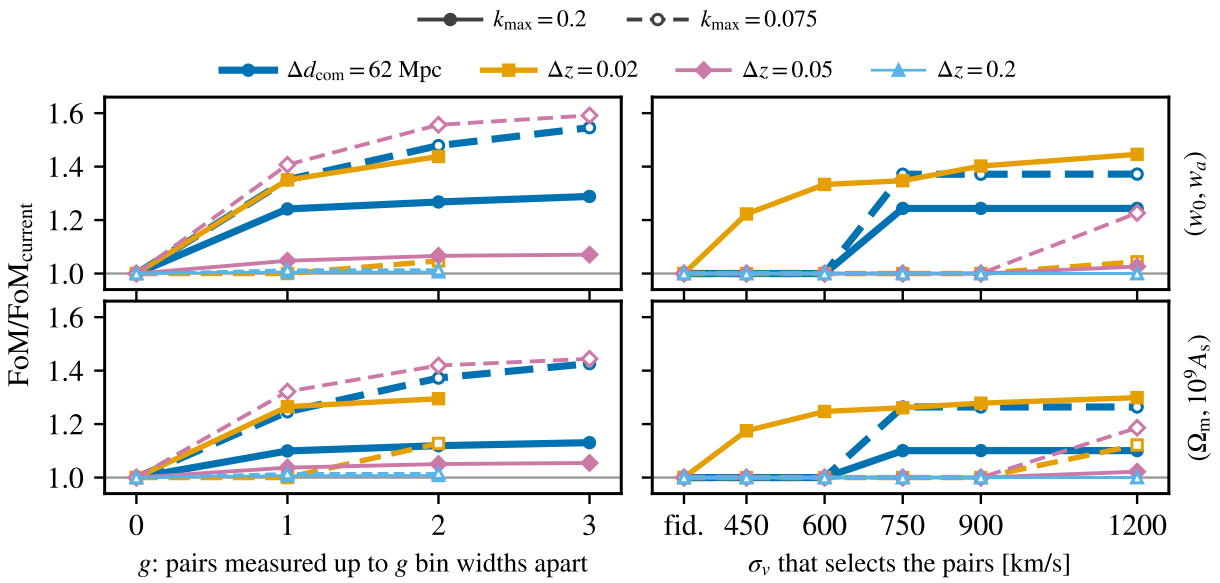


Figure 2. The figures of merit with pairs beyond the current data vector, over the current data vector's, with the dispersions fitted. Left column: the gap axis, pairs whose raw redshift ranges are at most g bin widths apart; right: the σ_v axis, the pairs selected by the overlap criterion with every dispersion at the common value s, the model kept at the fiducial dispersions ("fid.": the current selection). Top: FoM(w_0, w_a); bottom: FoM$(\Omega_m, 10^9 A_s)$; the panels share the y axis. Color, line width and marker: the layout; solid, filled markers: $k_{\max} = 0.2\,h\,\mathrm{Mpc}^{-1}$; dashed, open: 0.075; gray: one. $k_{\max}$ at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

through $b_{1,i}^2$ exactly as a larger clustering amplitude would, and a slope tilts the group's spectra against each other the way an evolving growth does – and how growth evolves with redshift is what w_0 and w_a control. The bias line and the growth history are degenerate: they imprint nearly the same pattern on the auto spectra, so the fit measures only a combination of them. The dispersions ride on the same degeneracy, since widening σ_v lowers an auto spectrum like a lower bias does, and like a change of w_0. A cross spectrum breaks the pattern: it carries the biases as the product $b_{1,i}\,b_{1,j}$, and its correlation coefficient with the two auto spectra, $C_\ell^{ij}/\sqrt{C_\ell^{ii}\,C_\ell^{jj}}$, is free of the linear biases at linear order. Each added pair is therefore one more measurement of the clustering two shells share that no bias line can reshape: it separates the bias line, and σ_v with it, from the growth history. Figure 3 reads the mechanism off the forecasts. With each group's level and slope held at the fiducial – the bias line fixed – most of the gain goes: at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ the factor 1.59 of $\Delta z = 0.05$ falls to 1.08, and the 1.55 of $\Delta d_{\rm com} = 62$ Mpc to 1.29; holding only the correlated residuals fixed instead leaves the gain almost unchanged. With every nuisance parameter fixed the added pairs keep at most 8% anywhere: the line-of-sight clustering information the added spectra carry by themselves is worth little once the biases are known.

The gain over wide bins. The ratios above compare each layout to its own current data vector; Figure 4 compares the layouts to each other, dividing every figure of merit by that of the wide $\Delta z = 0.2$ layout on the current data vector at the same cut. The thin slicing is worth far more than the added pairs: on the current data vector at $k_{\max} = 0.2\,h\,\mathrm{Mpc}^{-1}$ the thin layouts hold 26 ($\Delta z = 0.02$) and 22 ($\Delta d_{\rm com} = 62$ Mpc) times the wide layout's FoM(w_0, w_a), against 8 on $\Delta z = 0.05$, and the added pairs raise the first two factors to 37 and 28; at 0.075 the factors are 8,

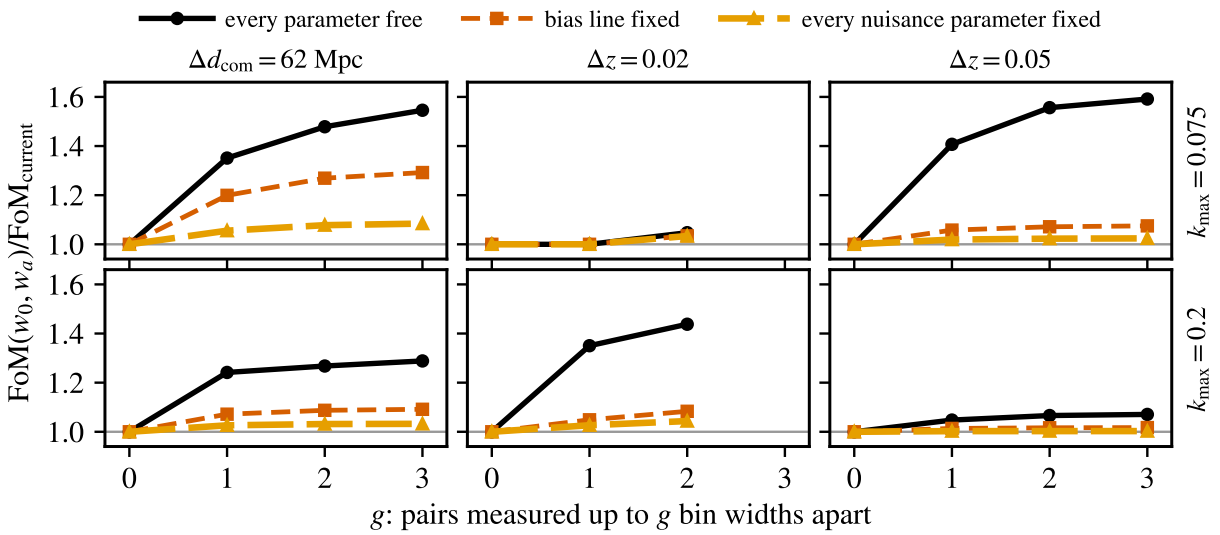


Figure 3. FoM(w_0, w_a) along the gap axis, over the current data vector's, with the dispersions fitted, on the three layouts whose gain is large. Black solid, filled circles: every parameter free; vermillion dashed, squares: each group's bias line – the level and the slope of $\beta_i = b_{1,i} D(\bar{z}_i)$ – held fixed; orange long-dashed, triangles: every nuisance parameter held fixed. Rows: $k_{\rm max} = 0.075$ and $0.2\,h\,{\rm Mpc}^{-1}$. $k_{\rm max}$ at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

13 and 4. Most of this advantage, too, comes from measuring each group's bias line: with the levels and slopes held fixed in both layouts, the thin layouts keep factors of about 7–9 at $0.2\,h\,{\rm Mpc}^{-1}$ in place of 22–26, because the thin slicing measures each group's bias evolution from its own spectra and the wide slicing, with one or two bins per group, cannot. The attribution against wide bins, block by block, is in `notes/cross_correlations`.

What is robust, and what is open. The gain is a property of the bias model, not of the residual prior. Three priors on the amplitudes pass the error-bar test of `notes/correlated_bias_prior_v2`, the test of Section 8 applied to each prior: this note's (the pessimistic prior), the baseline prior, whose residuals are smaller and correlated over a longer length, and the free prior, with no residual prior at all. All three leave every level and slope free, and the gain agrees across them within about 3% of the ratio. The growth-tied prior – one amplitude per group, with no slope and no residuals – fails that test, so no figure of merit is quoted for it. The analysis keeps the current data vector; the longer lists are a comparison against it. What stays open is a finite prior on the levels and slopes, which no passing prior imposes and which would keep only part of the gain. The templates, tables and checks behind these statements are in `notes/cross_correlations`, in its sections on what pairs beyond the list would add and on the free bias levels and slopes.

5 The bias model

The one-loop power spectrum. The galaxy power spectrum of bin i, in redshift space – with distances inferred from redshifts, so that peculiar velocities displace galaxies along the line of sight

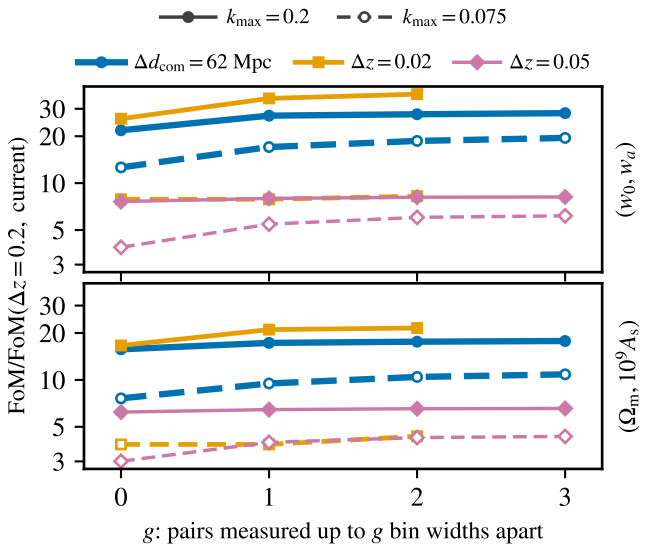


Figure 4. The figures of merit along the gap axis, over those of the $\Delta z = 0.2$ layout on the current data vector at the same cut, with the dispersions fitted ($\Delta z = 0.2$'s own gain along the axis stays below 2%; its ratio is not drawn). Top: FoM(w_0, w_a); bottom: FoM$(\Omega_m, 10^9 A_s)$; the panels share the y axis, logarithmic. Color, line width and marker: the layout; solid, filled markers: $k_{max} = 0.2\,h\,\mathrm{Mpc}^{-1}$; dashed, open: 0.075. k_{max} at $z = 0$; each bin is cut at $k_{max}/D(\bar{z}_i)$.

– is

$$\begin{aligned} P_{g,i}(k,\mu) \;&= b_{1,i}^2\, P_{mm}(k) + P_{loop}(k;\, b_{1,i}, b_{2,i}, b_{s^2,i}) + \alpha_{0,i}\, k^2 P_{mm}(k) \\ &\quad + \left(2\, b_{1,i} f\mu^2 + f^2\mu^4\right) P_{lin}(k) + \left(b_{1,i} + f\mu^2\right) f\, \alpha_{2,i}\, \mu^2 k^2 P_{lin}(k) \\ &\quad + \frac{1}{\bar{n}_i}\left(\mathrm{SN}_{0,i} + \mathrm{SN}_{2,i}\, f_{sat}\sigma_{sat}^2\, k^2\mu^2\right), \end{aligned} \tag{3}$$

with k the comoving wavenumber, μ the cosine of the angle to the line of sight, P_{mm} the nonlinear matter power spectrum (halofit, a fit to N-body simulations), P_{lin} the linear one, $f = d\ln D/d\ln a$ the growth rate and $\bar{n}_i$ the bin's number density. One-loop means one order beyond linear perturbation theory: corrections quadratic in the linear power spectrum. P_{loop} holds the one-loop terms of the quadratic bias b_2 and the tidal bias b_{s^2}, the responses of the galaxy density to the squared density contrast and to the squared tidal field [6, 7]; the third-order bias is zero. α_0 is the k^2 counterterm, which absorbs the sensitivity of the loops to small scales. The P_{lin} terms are linear (Kaiser) redshift-space distortions – the coherent infall toward overdensities, which boosts line-of-sight power by the factors $f\mu^2$ – on each bin's Fingers-of-God-convolved redshift distribution, with no redshift-space one-loop terms, and α_2 is the line-of-sight counterterm in the convention of the DESI 2024 full-shape analysis [8], in which $\alpha_2 k^2$ is a fractional correction to the velocity part of the Kaiser power. Equation (3) writes these terms with P_{lin}; their projection adds the Limber difference between P_{mm} and P_{lin} (Section 7), so where the Limber approximation holds they carry the nonlinear power. Both counterterms are in $h^{-2}\,\mathrm{Mpc}^2$. SN_0 is the white (scale-independent) noise beyond Poisson, in units of the Poisson shot-noise power $1/\bar{n}_i$ – the power the finite number of galaxies adds – that the covariance carries, and $\mathrm{SN}_2\, k^2\mu^2$ the stochastic term of the satellites' velocities, in units of $f_{sat}\sigma_{sat}^2/\bar{n}_i$, with f_{sat} the satellite fraction and σ_{sat} the satellites' velocity dispersion in distance units: $(0.1, 7.8\,h^{-1}\mathrm{Mpc})$ for LRG and $(0.1, 2.9\,h^{-1}\mathrm{Mpc})$ for ELG [8], and the LRG values for BGS.

The dispersions carry the line-of-sight counterterm. The model fixes $\alpha_2 = 0$ in every bin. On thin bins, widening a bin's Fingers-of-God dispersion σ_v – the width of the Gaussian that smooths its redshift distribution – damps the same line-of-sight power that α_2 corrects, and the two are nearly fully degenerate in these spectra. The model therefore samples the seven dispersions with Gaussian priors six times their base widths (the table of Section 2; the base widths, 145, 105, 70, 50, 50, 40, 30 km/s, are five times the median per-bin uncertainty of the dispersion measurements). With that widening, a fit without α_2 absorbs a line-of-sight correction of the DESI 2024 prior size through the dispersions alone, on every layout and at every cut; the degeneracy, the choice of the factor and the linear-response shifts behind it (parameter shifts to first order in the truth, as in Section 7) are in `notes/fog_parameter_degeneracy_study`.

One form for every bias quantity. Every bias quantity X of Eq. (3), in bin i of group g, is

$$X_i \;=\; X_g(\bar{z}_i) \;+\; t_{X,g}\,(\bar{z}_i - \bar{z}_g) \;+\; \varepsilon_{X,i}, \tag{4}$$

with $X_g(\bar{z}_i)$ the group value of X, which includes the calibration trend where X has one, $t_{X,g}$ a tilt in redshift across the group, $\bar{z}_g$ the mean redshift of the group's bins, and $\varepsilon_{X,i}$ a residual of the bin. The group value and the tilt are the freedom of the wide sample; the residual is what thin slicing may add within it (Section 6). Table 1 lists what each quantity carries.

For the linear bias the three parts enter as fractions of the group value $B_g/D(\bar{z}_i)$, which keeps the clustering amplitude $b_1 D$ constant across the group:

$$b_{1,i} \;=\; \frac{B_g}{D(\bar{z}_i)}\,\big(1+\delta_i\big), \qquad \delta_i \;=\; s_g\,(\bar{z}_i - \bar{z}_g) + \varepsilon_i, \tag{5}$$

with B_g the group amplitude, s_g the tilt, ε_i the residual and δ_i the fractional deviation from the group value. The group values of the quadratic and tidal biases are calibrations evaluated at the bin's own linear bias, plus a group offset:

$$b_{2,i} = b_2^{\rm L}(b_{1,i}) + \Delta_{2,g} + t_{2,g}\,(\bar{z}_i - \bar{z}_g), \qquad b_{s^2,i} = -\tfrac{4}{7}\,\big(b_{1,i}-1\big) + \Delta_{s,g} + t_{s,g}\,(\bar{z}_i - \bar{z}_g), \tag{6}$$

with $b_2^{\rm L}(b_1) = 0.412 - 2.143\,b_1 + 0.929\,b_1^2 + 0.008\,b_1^3$ the separate-universe fit of Lazeyras et al. [9] and the second relation the co-evolution of a population formed with local bias [7]. Only the linear bias carries a residual of its own: the quadratic and tidal biases receive each bin's residual through $b_{1,i}$ in their calibrations, and the counterterms and stochastic terms have none.

Why SN_2 matters on thin bins. A constant power N projects exactly as $N\int d\chi\, W_i W_j/\chi^2$ onto bins i and j, with χ the comoving distance and $W_i(\chi)$ the Fingers-of-God-convolved kernel of bin i; bins of two different samples share no stochastic term. A power proportional to $k^2\mu^2 = k_\parallel^2$ is that of the line-of-sight derivative of a white noise, so SN_2 projects with each kernel replaced by its derivative, $W_i' = dW_i/d\chi$:

$$C_\ell^{ij}\big|_{\mathrm{SN}_2} \;=\; U_{ij}\int d\chi\,\frac{W_i'(\chi)\,W_j'(\chi)}{\chi^2}, \tag{7}$$

with U_{ij} the mean of the two bins' amplitudes $\mathrm{SN}_2 f_{\rm sat}\sigma_{\rm sat}^2/\bar{n}$. A thin bin's W_i' peaks at its two edges with opposite signs, and adjacent bins share an edge, where one kernel rises as the other falls: the projected SN_2 of neighbors is anticorrelated. Normalized by the two auto terms, the adjacent term has a median of -0.50 over the 73 adjacent pairs of one sample on $\Delta d_{\rm com} = 62$ Mpc, against $+0.05$ for SN_0. No white noise, per group or per bin, mimics this pattern, and a fit without SN_2 is biased on the thin layouts already at $k_{\rm max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ (`notes/higher_order_bias`).

quantity	group value	tilt	residual
b_1	$B_g/D(\bar{z}_i)$; B_g flat	s_g, flat	ε_i, Eq. (8)
b_2	$b_2^{\rm L}(b_{1,i}) + \Delta_{2,g}$; $\Delta_{2,g}$ width 2	flat	none of its own
b_{s^2}	$-\frac{4}{7}(b_{1,i}-1) + \Delta_{s,g}$; $\Delta_{s,g}$ width 1	flat	none of its own
α_0	width $0.5/k_{\rm max}^2$	flat	none
α_2	fixed to 0	none	none
SN_0	width 2	none	none
SN_2	width 5	none	none

Table 1. What each bias quantity carries in Eq. (4). A width is that of a Gaussian prior centered on zero, so the fiducial model sits on the calibrations with no counterterm and no stochastic term beyond Poisson; flat marks a flat prior. The α_0 width is the rule of the DESI 2024 full-shape analysis [10, 8], a correction of 0.5 at the cut, evaluated at each cut ($12.5\,h^{-2}\,\mathrm{Mpc}^2$ at $0.2\,h\,\mathrm{Mpc}^{-1}$). α_2 is fixed: the widened dispersion priors carry its freedom. A group of one bin has no tilt.

6 The prior: the freedom of the wide sample

The principle. DESI selects each group as one wide sample, and what is known about its bias, from calibrations to the priors of full-shape analyses, refers to that sample. Slicing it into thin bins adds measurements of the clustering, not knowledge of the bias. A thin-bin prior is therefore fair only if it leaves each group the freedom of its wide sample, whatever the slicing. In Eq. (4) that freedom is the group value and the tilt, whose priors are set for the wide sample; the residual is what the slicing may add within it, local features of the size the selection can produce, and its prior must not tighten as the bins get thinner.

The linear bias. The level B_g and the tilt s_g are free, and the residual has a Gaussian prior, zero across groups and correlated within one,

$$\langle \varepsilon_i\, \varepsilon_j \rangle \;=\; \sigma_g^2\, \exp\!\left(-\,\frac{|\bar{z}_i - \bar{z}_j|}{L}\right), \tag{8}$$

with σ_g the width of one bin's residual and L the correlation length in redshift. The kernel defines a random process $\varepsilon(z)$ along each group before any binning, which a layout only samples at its bin redshifts: the prior on a coarse set of redshifts is the marginal of the prior on a finer one – integrate the finer prior over the bins the coarse layout does not use and the coarse prior remains – so slicing finer adds no prior knowledge. Two neighbors at separation Δz differ by an rms $\sigma_g[2(1-e^{-\Delta z/L})]^{1/2}$. The prior charges only the residual around each group's best straight line (Section 7), so a group of one or two bins, as on $\Delta z = 0.2$, has a free bias in every bin.

No measurement resolves the bias of these samples below $\Delta z \approx 0.1$, so the sizes come from what selects them (`notes/correlated_bias_prior_v2`). Coherent changes move a whole group along a nearly straight line, which the free tilt absorbs: a conserved population, whose bias obeys $b_1 - 1 \propto 1/D$ [11], departs from B_g/D by about 1% per $\Delta z = 0.05$, and the $r < 19.5$ flux limit of BGS [12] drifts its bias by 22% across the group. Local changes, such as the 4000 Å break – a sharp step in the spectrum of an old stellar population – crossing an edge of a selection filter over $\Delta z \approx 0.03$–0.05, move bins independently. Hence $\sigma_g = 3\%$ in every group, above the largest derived departure from a line (the 2.4% rms that the BGS flux limit leaves), and $L = 0.05$, the scale of a filter-edge crossing; neighbors at $\Delta z = 0.02$ then differ by 2.4% rms. The draws of Figure 5 keep the line of the known drift and add local features of a few percent.

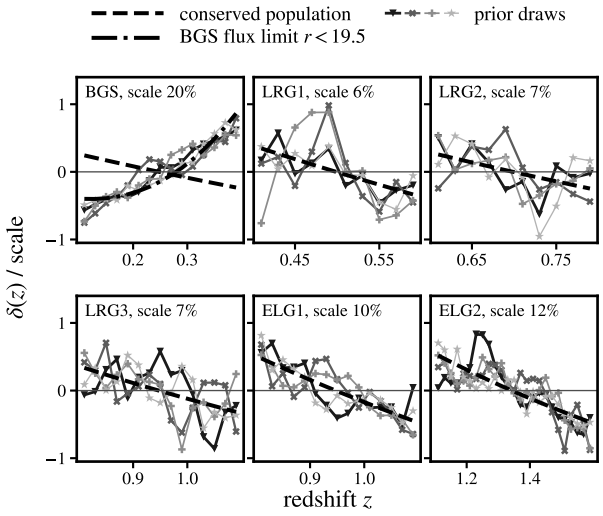


Figure 5. Four draws of the prior on the $\Delta z = 0.02$ bins, one panel per group, each panel divided by the round scale written in it. Thick black dashed: the conserved-population drift; thick black dash-dot (BGS only): the $r < 19.5$ flux-limit drift. Thin gray lines: the draws of $\delta(z)$, each with its own shade, line width and marker. Every draw keeps the best straight line through its group's reference drift, the flux limit in BGS and the conserved population elsewhere, and differs from it by a residual drawn with $\sigma_g = 3\%$ and $L = 0.05$. The level of every curve is removed, since B_g is free.

The higher-order terms. The calibrations of b_2 and b_{s^2} describe halo populations; a wide sample adds how far it sits from them. Each group therefore gets one offset per family, with the widths of Table 1: twice what the known imperfection of the calibrations requires (the scatter of the Lazeyras fit and the 30% spread between co-evolution and excursion-set tidal biases), so that the calibrations act as prior centers where the loops reach 15% of the galaxy power. A calibration of the wide sample says nothing about how the offset changes inside it, so the tilt is free; it also absorbs exactly a calibration that drifts linearly across a group. The counterterm α_0 follows the same pattern, since its scale follows the host halos of the selection. The stochastic amplitudes do not tilt: their Poisson part is known bin by bin.

7 Projection, covariance and the Fisher forecast

Non-Limber spectra. The Limber approximation evaluates the power spectrum only at the transverse wavenumber $(\ell + 1/2)/\chi$: accurate for a wide radial kernel, which averages the line-of-sight modes away, it misses the modes that a thin bin with sharp edges passes. The exact projection keeps them: between bins i and j, with $W_i(\chi) = n_i(z)\, dz/d\chi$ the radial kernel of bin i, n_i its normalized redshift distribution, j_ℓ the spherical Bessel function and $P(k;\, z_1, z_2)$ the unequal-time power spectrum of the projected field at $z_1 = z(\chi_1)$ and $z_2 = z(\chi_2)$,

$$C_\ell^{ij} \;=\; \frac{2}{\pi} \int d\chi_1\, W_i(\chi_1) \int d\chi_2\, W_j(\chi_2) \int_0^\infty dk\, k^2\, P(k;\, z_1, z_2)\, j_\ell(k\chi_1)\, j_\ell(k\chi_2). \tag{9}$$

The double Bessel integral is the cost: $j_\ell(k\chi_1)\, j_\ell(k\chi_2)$ oscillates and beats, and quadrature over it is slow and unstable.

Fang et al. [2] (FKEM) remove that cost by splitting the spectrum into its linear part and a nonlinear remainder. The linear part is separable, $P_{\rm lin}(k;\, z_1, z_2) = D(z_1)\, D(z_2)\, P_{\rm lin}(k)$, with $P_{\rm lin}(k)$ the linear matter power spectrum today, so in Eq. (9) each χ integral closes around its own Bessel function and becomes a one-dimensional transform that the FFTLog algorithm, a fast Fourier transform on a logarithmic grid, computes quickly. The remainder $P_{\rm mm} - P_{\rm lin}$, with $P_{\rm mm}$ the nonlinear matter power spectrum, matters only at high k, which projects to multipoles where the Limber approximation is accurate, so it is added in Limber:

$$C_\ell \;=\; C_\ell^{\rm exact}[P_{\rm lin}] \;+\; C_\ell^{\rm Limber}[P_{\rm mm}] \;-\; C_\ell^{\rm Limber}[P_{\rm lin}]. \tag{10}$$

FKEM projects three transfer kernels per galaxy bin: density, redshift space and magnification. The density and redshift-space (Kaiser) parts together give, with $b_{1,i}$ the bin's linear galaxy bias and $f = d\ln D/d\ln a$ the linear growth rate (a the scale factor),

$$\Delta_\ell^i(k) \;=\; \int d\chi\, W_i(\chi)\, D(z)\, \big[\, b_{1,i}\, j_\ell(k\chi) \;-\; f(z)\, j_\ell''(k\chi) \,\big] : \tag{11}$$

each factor $f\mu^2$ of the flat-sky Kaiser power, with μ the cosine of the angle to the line of sight, becomes a $-j_\ell''$ in the exact projection. In the Limber terms of Eq. (10) the identity $j_\ell''(x) = [\ell(\ell-1)/x^2 - 1]\, j_\ell(x) + (2/x)\, j_{\ell+1}(x)$ writes the second derivative with the orders ℓ and $\ell+1$, and Limber evaluates each order at its own distance, $\chi_\ell = (\ell+1/2)/k$ and $\chi_{\ell+1} = (\ell+3/2)/k$, so the Limber redshift-space kernel combines the bin's radial kernel at two neighbouring distances: redshift-space distortions move galaxies between neighbouring shells. The third kernel, lensing magnification, is proportional to the magnification bias $b_{\rm mag}$, the response of the observed number density to the lensing convergence; this forecast sets $b_{\rm mag} = 0$ in every DESI bin and every LSST Y1 lens bin, so magnification is neglected.

For thin bins the split of Eq. (10) is not enough. Limber's error shrinks with the width of the radial kernel, so for a thin bin the Limber remainder stays wrong far past the switch multipole: the hand-off fails exactly where accuracy was the goal. The density–density part of every DESI spectrum – the linear-bias term, the one-loop terms and the α_0 counterterm of Eq. (3) – is therefore projected by a low-rank factorization instead. Per bin, a singular value decomposition writes the square root of the real-space galaxy power spectrum $P_g(k,z)$ as a sum of five products of a function of k and a function of z, each projected with its own FFTLog transform; the unequal-time spectrum is the geometric mean $\sqrt{P_g(k,z_1)}\,\sqrt{P_g(k,z_2)}$, exact at equal times and treating the two fields as fully correlated otherwise. This carries the full one-loop power with no Limber hand-off up to a multipole cap at $\ell = 1000$, below which every kept band lies.

The hybrid in one sentence: the density–density part runs through the low-rank factorization; the Kaiser terms and the α_2 counterterm keep the FKEM split, projected exactly with $P_{\rm lin}$ and corrected in Limber by Eq. (10); the stochastic terms are projected as exact templates; and magnification is neglected. The one-loop power is not pushed through the velocity terms because velocities do not grow like the density field once nonlinear.

The covariance. The band-power covariance is Gaussian, computed at the one-loop fiducial (every bin's linear bias on its group's trend, b_2 and b_{s^2} on their calibrations, every other bias term zero) for 14,000 square degrees, with its inverse rescaled to the forecast's 12,300. The data vectors are the model's own prediction at that fiducial.

The Fisher matrix. Let $\mathbf{m}(\mathbf{p})$ be the model band powers at the parameters $\mathbf{p}$, $\mathbf{p}_0$ the fiducial point, C the band-power covariance, and $\mathsf{D} = \partial\mathbf{m}/\partial\mathbf{p}$ at $\mathbf{p}_0$ (sans-serif, unlike the growth factor D). At a scale cut that keeps the set K of band powers, the Fisher matrix of a fit is

$$F \;=\; \mathsf{D}_K^{\mathsf{T}}\, \mathsf{C}_{KK}^{-1}\, \mathsf{D}_K \;+\; F_{\rm prior}, \tag{12}$$

with D_K the kept rows of D and C_{KK} the kept block of C. $F_{\rm prior}$ adds $1/\sigma_{\rm pr}^2$ to the diagonal for a Gaussian prior of width $\sigma_{\rm pr}$ and nothing for a flat one. For each group and each quantity that carries a residual (Table 1) it adds the precision $\mathsf{P} = \mathsf{Q} - \mathsf{Q}\mathsf{U}(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\mathsf{U})^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q}$ on the bins' deviations from the group value (δ_i for b_1), with Q the inverse kernel matrix of Eq. (8) and U the matrix with columns 1 and $\bar z_i - \bar z_g$: it charges only the residual around the group's best straight line (the projected precision of `notes/correlated_bias_prior_v2`). The marginalized error of parameter j

is $\sigma_j = [(F^{-1})_{jj}]^{1/2}$, and the FoMs of Section 3 are $[\det \mathsf{C}_p]^{-1/2}$, with C_p the 2×2 block of F^{-1} of the two parameters. Where a cut leaves a combination of nuisance parameters without information, for instance the flat tilts of a group most of whose bins have lost their bands, F is inverted as a pseudo-inverse: scaled to unit diagonal, diagonalized, and restricted to the eigen-directions above 10^{-12} of the largest eigenvalue. A dropped direction may not touch (w_0, w_a), Ω_m or A_s by more than 10^{-6}; otherwise the cut gives no fit.

Derivatives. The columns of D come from three kinds of derivatives at the fiducial point.

1. The cosmological parameters, the σ_v and the per-bin $b_{1,i}$, $b_{2,i}$, $b_{s^2,i}$ and $\alpha_{0,i}$, on which the band powers depend nonlinearly, take the five-point central stencil

$$\frac{\partial \mathbf{m}}{\partial \theta} \simeq \frac{\mathbf{m}(\theta_0 - 2\epsilon) - 8\,\mathbf{m}(\theta_0 - \epsilon) + 8\,\mathbf{m}(\theta_0 + \epsilon) - \mathbf{m}(\theta_0 + 2\epsilon)}{12\,\epsilon}, \tag{13}$$

   with θ_0 the fiducial value and $\epsilon = h\,\theta_0$, $h = 0.02$ (an absolute step where $\theta_0 = 0$); its truncation error is of order ϵ^4. Every evaluation is a full call of the likelihood, CAMB included.

2. SN_0, SN_2 and α_2 enter linearly, so their columns are exact: the projected templates per unit amplitude for the stochastic terms (Eq. (7) for SN_2), and a two-point difference for α_2.

3. The per-bin columns are computed in batches. A spectrum C_ℓ^{ij} depends on the biases of bins i and j only, so the bins of a set $\mathcal{S}$ in which no two bins share a measured spectrum can be stepped together. With $\partial_{\mathcal{S}}\mathbf{m}$ the stencil of Eq. (13) with the bias x stepped in every bin of $\mathcal{S}$,

$$\frac{\partial m_r}{\partial x_i} = \begin{cases} (\partial_{\mathcal{S}}\mathbf{m})_r & \text{if band power } r \text{ belongs to a spectrum that contains bin } i \in \mathcal{S}, \\ 0 & \text{otherwise.} \end{cases} \tag{14}$$

   A greedy coloring of the graph of measured pairs – bins are the vertices, measured pairs the edges, and bins of one color share no edge – gives four such batches on every layout, so each per-bin family costs 16 evaluations (four batches times the four off-center points of Eq. (13)), whatever the number of bins.

One base, every model. The columns of one layout, for the cosmology, the σ_v and the seven per-bin families b_1, b_2, b_{s^2}, α_0, α_2, SN_0 and SN_2, form a base matrix B, computed once and cached. A model is a fixed linear map from its parameters p_j to the base quantities κ_a,

$$\mathsf{D} = \mathsf{B}\,\mathsf{M}, \qquad \mathsf{M}_{aj} = \frac{\partial \kappa_a}{\partial p_j}\,: \tag{15}$$

a group offset puts ones on its group's bins in its family's block, a tilt the weights $\bar{z}_i - \bar{z}_g$, a bin's linear bias the chain rule through the calibrations of Eq. (6), and the untied versions of Section 9 per-bin identities. With $\mathsf{G}_K = \mathsf{B}_K^{\mathsf{T}} \mathsf{C}_{KK}^{-1} \mathsf{B}_K$ computed once per cut, every model and prior at that cut has $F = \mathsf{M}^{\mathsf{T}} \mathsf{G}_K \mathsf{M} + F_{\mathrm{prior}}$: a change of model, prior, cut or truth needs no new evaluation of the data vector.

Shifts and the error-bar test. A truth that departs from the fitted model moves the band powers by $\Delta\mathbf{m}$, and the best fit, to first order, by

$$\delta\mathbf{p} \;=\; F^{-1}\,\mathsf{D}_K^{\mathsf{T}}\,\mathsf{C}_{KK}^{-1}\,\Delta\mathbf{m}_K, \tag{16}$$

exactly for the terms linear in their amplitude (α_2, SN_0, SN_2) and to first order for the biases, on which the band powers depend quadratically. Because $\delta\mathbf{p}$ is linear in the truth, the average behind the error-bar test of Section 8 is exact in closed form; for (w_0, w_a),

$$\langle\chi^2\rangle \;=\; \Delta^2_{\mathrm{drift}} \;+\; \mathrm{tr}\big[\mathsf{C}_p^{-1}\big(\mathsf{A}\,\Sigma\,\mathsf{A}^{\mathsf{T}} + \mathsf{N}_p\big)\big], \tag{17}$$

with Δ_{drift} the shift of the known drift alone in units of the ellipse, A the response of (w_0, w_a) to the bins' deviations δ_i, Σ the covariance of the residuals drawn from the prior, and N_p the (w_0, w_a) block of $F^{-1}F_0F^{-1}$, the share of the data noise and the other priors, F_0 being F without the correlated prior. For Ω_m or A_s the same holds with the parameter's own row.

Numerical convergence. Noise in the data vector, from its integration grids and CAMB's tolerances, enters a finite difference divided by the step. A column $\mathbf{d}_j + \mathbf{e}_j$ with noise $\mathbf{e}_j$ raises F_{jj} on average by $\langle\mathbf{e}_j^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{e}_j\rangle > 0$: noise reads as information and inflates the FoM. Settings that converge a χ^2 to the standard of MCMC runs, $|\Delta\chi^2| < 0.2$, are therefore not enough. On $\Delta z = 0.05$ they inflate $\mathrm{FoM}(w_0, w_a)$ by 56% at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ and 13% at $0.2\,h\,\mathrm{Mpc}^{-1}$; the excess comes from noise in the w_0, w_a and Ω_m columns. That spurious information also weakens the degeneracies the error bars rest on, such as the one between the dispersions and the cosmology, which is why the final settings are needed. The criterion of convergence is step independence: the FoMs, the errors and the shifts must not move when the stencil step changes or the numerical controls are refined further. A study on $\Delta z = 0.05$ (Figure 6) starts from a configuration with every control raised: the grid factor $g = 128$, where g counts the comoving distances of the non-Limber integrals in units of 511; CosmoLike's `accuracyboost` 4 and `integration_accuracy` 2; the low-rank engine's rank 8 and multipole cap 2000; and CAMB's `AccuracyBoost` 2, `k_per_logint` 30 and maximum wavenumber 80 Mpc^{-1}. From there it changes one control at a time and compares each configuration with the converged value, the five-point stencil at step h on $g = 512$. The final settings are:

- the final grid $g = 256$, eight times the MCMC grid of $\Delta z = 0.05$, with CosmoLike's `accuracyboost` 3 and `integration_accuracy` 1;
- CAMB's `AccuracyBoost` 2.0, `k_per_logint` 30 and a maximum wavenumber of 60 Mpc^{-1}, with halofit's bisection tolerance tightened from 10^{-3} to 10^{-7} (the halofit tolerance patch);
- five-point stencils for every stencil and per-bin column.

Against the converged value they give $\mathrm{FoM}(w_0, w_a)$ +2.1% at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ and −0.5% at $0.2\,h\,\mathrm{Mpc}^{-1}$ ($\mathrm{FoM}(\Omega_m, 10^9 A_s)$ +1.8% and −0.4%), marginalized errors within 1.6% and shifts within 0.007σ. From step $h/2$ to $3h$ both FoMs stay within 1.6% on $\Delta z = 0.05$, and from $h/2$ to $2h$ within 0.5% on $\Delta d_{\mathrm{com}} = 62$ Mpc. Two cross-checks agree. DerivKit, which fits a polynomial through at least seven evaluations around the fiducial point, matches the stencils within 2.1%. The study's route, which unties the per-bin parameters in the likelihood, matches the production route, which steps them through CosmoLike's setter: exactly for b_2, b_{s^2} and α_0, within 5×10^{-11} for b_1 and within 5×10^{-14} for the stencil columns.

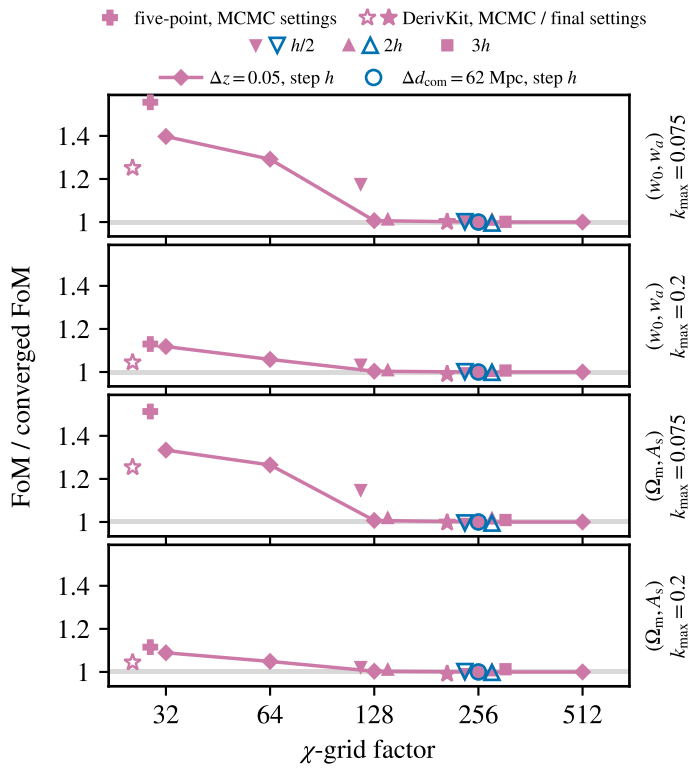


Figure 6. Convergence of the forecast on $\Delta z = 0.05$: the FoMs of the model divided by their converged values (five-point stencils at step h on the grid factor $g = 512$). From the top, FoM(w_0, w_a) and FoM$(\Omega_m, 10^9 A_s)$ at $k_{\max} = 0.075$ and $0.2\,h\,\mathrm{Mpc}^{-1}$. Diamonds on the line: the scan in g at step h, which holds every other control at the study's starting configuration ($g = 128$ with the other controls raised; Section 7); $g = 256$ is the final setting. Filled down-triangles, up-triangles and squares beside a grid: steps $h/2$, $2h$ and $3h$. Plus: five-point stencils at the MCMC settings. Open and filled stars: DerivKit at the MCMC and at the final settings. Open blue markers: $\Delta d_{\mathrm{com}} = 62$ Mpc at the final settings, steps $h/2$ (down-triangle) and $2h$ (up-triangle) against its own step-h value (ring). Gray band: $1 \pm 1\%$. $k_{\max}$ at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

Cost. A layout's base takes 154 evaluations of the data vector, about 20 to 40 s each at the final settings, so one to two hours per thin layout; every model, prior, cut and truth then costs a fraction of a second. `notes/fisher_machinery` gives the details.

8 Forecast

Figures of merit. Figure 7 draws both FoMs against $k_{\max}$. FoM(w_0, w_a) is 0.564, 0.352, 0.173 and 0.0447 at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ and 6.70, 7.95, 2.34 and 0.307 at $0.2\,h\,\mathrm{Mpc}^{-1}$, in layout order ($\Delta d_{\mathrm{com}} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2); FoM$(\Omega_m, 10^9 A_s)$ at $0.2\,h\,\mathrm{Mpc}^{-1}$ is 50.3, 53.2, 19.9 and 3.20. The thin layouts carry most of the information. LSST Y1 3x2pt alone gives 17.3 and 215; DESI C_ℓ^{gg} alone stays below it in both FoMs at every cut.

The error bars are honest. How well a fit's quoted error bars cover where the fit actually lands is measured by $\eta = \sqrt{\langle \chi^2 \rangle / n_{\mathrm{dof}}}$, with $\chi^2 = \Delta p^{\mathsf{T}} \mathsf{C}_p^{-1} \Delta p$, Δp the recovered minus the true parameters, C_p the fit's marginalized covariance, $n_{\mathrm{dof}} = 2$ for (w_0, w_a) and 1 for Ω_m or A_s, and $\langle \cdot \rangle$ the average over the truths and the data noise; $\eta = 1$ for honest error bars. Over random bias histories, the known drifts plus a residual drawn from the prior, $\eta \leq 1.030$ at every cut and on every layout (at most 1.019 for (w_0, w_a), 1.030 for Ω_m and 1.004 for A_s), and the 1σ region covers the truth in at least 66.8% of universes, against 68.3% for honest error bars.

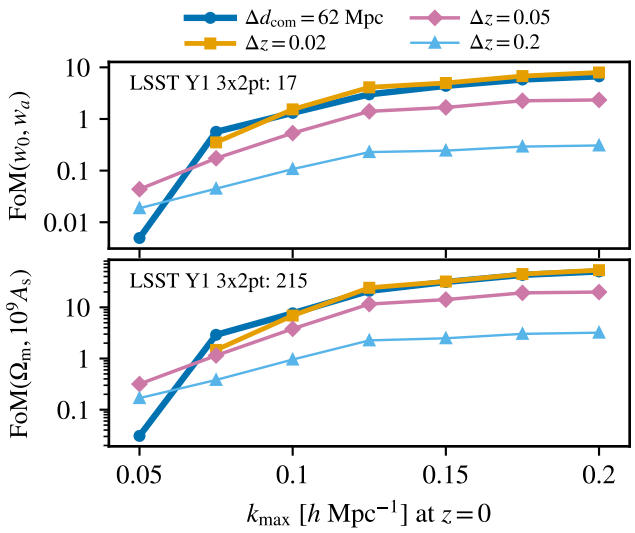


Figure 7. FoM(w_0, w_a) (top) and FoM$(\Omega_m, 10^9 A_s)$ (bottom) of DESI C_ℓ^{gg} alone against $k_{\rm max}$. Color, line width and marker: the layouts. Text in each panel: LSST Y1 3x2pt alone. On $\Delta z = 0.02$ the cut at 0.05 keeps no band. $k_{\rm max}$ at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

9 What the grouping is worth

The principle of Section 6 says that a thin-bin prior must keep the freedom of the wide original samples, and add nothing to it. Figure 8 puts the principle to a test on six versions of the priors: five untie the bias bin by bin, and one ties it harder, to a constant across each sample. The top two rows divide each version's FoM by the model's; the bottom row asks whether its error bars are honest, with the statistic η of Section 8 for (w_0, w_a) over the same random bias histories. The group width of a parameter is the width the model gives it (Table 1), and N is the number of bins of its group. The first four versions free $b_{1,i}$ in every bin and give Δ_2, Δ_s, α_0, SN_0 and SN_2 one value per bin; the fifth keeps the model whole, its b_1 prior included; the sixth ties every bin of a sample to one value. Every version keeps the model's dispersion priors, with no α_2.

Why a naive per-bin prior gains. Give each of N bins its own offset, independent of the others, each with a Gaussian prior of width σ. Their average is the sum of the N offsets divided by N. Variances of independent numbers add, so the sum has variance $N\sigma^2$ and the average $N\sigma^2/N^2 = \sigma^2/N$: the average has width $\sigma/\sqrt{N}$. With $\sigma = 2$, the group width of Δ_2, and $N = 9$, the prior pins the group's average offset to $2/3$, three times tighter than the wide sample knows it. Slice the same sample into more bins and the pin tightens further, with no new knowledge of the bias.

The six versions, in the words of the figure's legend:

- **per bin, group width**: looks looser than the model, but pins each group's average to the group width over $\sqrt{N}$. It lies above the model on every layout and at every cut, by up to a factor of 1.54: a gain by accident, from the slicing.
- **per bin, group width** $\times\sqrt{N}$: gives each group's average back the group width. On $\Delta z = 0.2$, whose groups hold one or two bins, it still gains, because it fixes the trend inside a group that the model leaves free.
- **per bin,** $\times\sqrt{N}$ **+ group tilt**: frees that trend too, and so restores the whole wide-sample freedom. Its prior is at least as loose as the model's in every direction (each bin's variance is

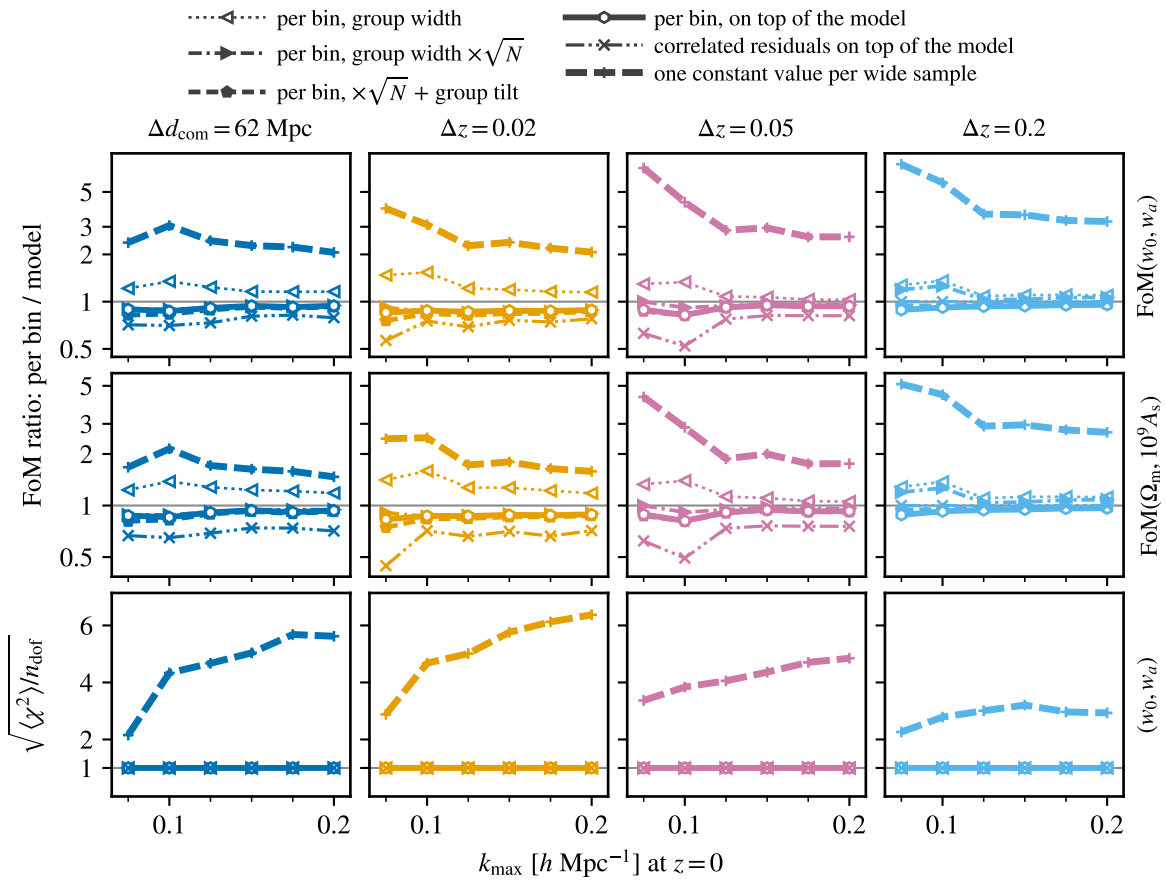


Figure 8. Six versions of the bias priors against the model, one column per layout. Top and middle rows: FoM(w_0, w_a) and FoM$(\Omega_m, 10^9 A_s)$, each divided by the model's. Bottom row: $\eta = \sqrt{\langle\chi^2\rangle/n_{\rm dof}}$ of (w_0, w_a) over the random bias histories of Section 8; for the constant version the truth also holds the bias's own evolution inside each sample. The group width is the width the model gives a group parameter, and N the number of bins of the group. The first four versions free $b_{1,i}$ in each bin and give Δ_2, Δ_s, α_0, SN$_0$ and SN$_2$ one value per bin. Dotted, open left triangles: the group width; dash-dot, filled right triangles: the group width times $\sqrt{N}$; dashed, filled pentagons: the same plus a free tilt per group; solid, open hexagons: the group width on top of the model's offsets and tilts. The line width grows in that order. Thin dash-dot-dot line, crosses: the model, its b_1 prior included, plus correlated per-bin residuals on Δ_2, Δ_s and α_0. Thick long-dashed line, plus signs: one constant value of every bias quantity per wide sample. Thin gray line: a ratio of 1, and honest error bars in the bottom row. There the other five versions coincide at 1; the solid line, drawn on top, hides all but the crosses. The cuts start at $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$: at $0.05\,h\,{\rm Mpc}^{-1}$ the model's own FoM(w_0, w_a) is below 0.05. $k_{\rm max}$ at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

N times the squared group width, against one squared group width for the shift common to all bins), so it can only lose FoM.

- **per bin, on top of the model**: keeps the model's group offsets and tilts, the wide-sample freedom, and adds an independent per-bin value of the group width to every term.

- **correlated residuals on top of the model**: keeps the model exactly and adds to each of Δ_2, Δ_s and α_0 a per-bin residual with the prior of b_1's: Gaussian, of the term's group width, correlated in redshift over $L = 0.05$, and charged only on the deviations from the group's line. In the form of Eq. (4) it fills the residual column of Table 1 for these three terms. In a group

of one or two bins the offset and the tilt already fix every bin, so on $\Delta z = 0.2$ this version is the model.

- **one constant value per wide sample**: every thin bin of a group shares one value of every bias quantity, constant in redshift. The linear bias is a plain constant, $b_{1,i} = B_g$ instead of $B_g/D(\bar{z}_i)$, and Δ_2, Δ_s, α_0, SN_0 and SN_2 take one value per group, under the model's priors on the group values, with no tilts and no residuals. It is the simplest way to slice a wide sample.

What untying the bins costs. The last two versions keep the wide-sample freedom and add only the freedom of untied bins, so they measure what untying really costs, and they differ sharply. Independent per-bin terms on top of the model cost a factor of 1.12 to 1.17 in $\mathrm{FoM}(w_0, w_a)$ on the thin layouts at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$, and 1.06 to 1.13 on the three thinner layouts at $0.2\,h\,\mathrm{Mpc}^{-1}$. Correlated residuals cost a factor of 1.4 to 1.8 on the three thinner layouts at $0.075\,h\,\mathrm{Mpc}^{-1}$, the most on $\Delta z = 0.02$, and 1.23 to 1.29 on the same layouts at $0.2\,h\,\mathrm{Mpc}^{-1}$. A residual that is smooth in redshift can mimic a change of the cosmology across neighboring bins, while independent per-bin terms average down over the bins of a group. On $\Delta z = 0.2$ neither version costs more than a factor of 1.13 at any cut.

What a constant per sample claims. The sixth version claims the largest gains of the figure: 2.4 to 7.5 times the model's $\mathrm{FoM}(w_0, w_a)$ over the layouts at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$, 2.1 to 3.2 at $0.2\,h\,\mathrm{Mpc}^{-1}$, and 1.5 to 5.1 times $\mathrm{FoM}(\Omega_m, 10^9 A_s)$. The bottom row prices the claim. For this version the truth also holds the evolution of the bias inside each sample, which a constant cannot follow: the fiducial bias itself evolves, the linear bias as $B_g/D(\bar{z}_i)$, and b_2 and b_{s^2} with it through the calibrations. That evolution alone shifts (w_0, w_a) by 3.8 to 6.0σ at $0.075\,h\,\mathrm{Mpc}^{-1}$ and 4.7 to 17σ at $0.2\,h\,\mathrm{Mpc}^{-1}$; the known drifts partly cancel it, which is why η comes out smaller than those shifts. The error bars are still 2.2 to 3.4 times too small at $0.075\,h\,\mathrm{Mpc}^{-1}$ and 2.9 to 6.4 at $0.2\,h\,\mathrm{Mpc}^{-1}$, where they reach 3.0 for Ω_m and 4.7 for A_s, and the 68.3% region covers the truth in 5% to 11% of universes (5.0% at the worst, on $\Delta z = 0.02$). The gain is the information the evolution of the bias would have to supply, asserted instead by the prior: where the model keeps the freedom the bias has on the wide original samples, a constant removes even the freedom the wide sample's own evolution needs. Even $\Delta z = 0.2$, one or two bins per sample, gains a factor of 3.2 to 7.5 and fails the test: giving up the tilts and the residuals there gives up real freedom.

What the bottom row tests. The random truths change only the linear bias, so the bottom row tests the linear bias, not the higher-order terms. The four versions that free $b_{1,i}$ per bin sit exactly at 1 by construction: a free per-bin linear bias absorbs any change of the linear bias, so their honesty there says nothing about their higher-order freedom. The version with correlated residuals, whose b_1 keeps the model's prior, stays between 0.99 and 1.01, and the model itself at most 1.019 (Section 8).

10 Where the details are

- `notes/correlated_bias_prior_v2`: the linear-bias prior and its physics, the projected precision, the random bias histories and the error-bar test, and the comparisons of priors and models.

- `notes/higher_order_bias`: the one-loop model, its calibrations, the size and exact projection of every term, and its validation.
- `notes/linear_bias_kmax`: the reach of linear bias, which motivates the one-loop terms.
- `notes/fisher_machinery`: the Fisher derivatives, their convergence study, and the assembly of every Fisher matrix, shift and $\langle\chi^2\rangle$.
- `notes/cross_correlations`: which thin-bin pairs the data vector measures, the Fingers-of-God blur, and the study of pairs beyond the list – the gap and σ_v axes, the attribution of the gain to the free bias levels and slopes, and the comparison across the linear-bias priors.
- `notes/fog_parameter_degeneracy_study`: the degeneracy of the line-of-sight counterterm α_2 with the dispersions σ_v, the choice of their prior widths, and the linear-response shifts that estimate how safe a chain run with this model would be.

References

[1] J. Asorey, M. Crocce, E. Gaztañaga and A. Lewis, "Recovering 3D clustering information with angular correlations", 2012, arXiv:1207.6487.

[2] X. Fang, E. Krause, T. Eifler and N. MacCrann, "Beyond Limber: efficient computation of angular power spectra for galaxy clustering and weak lensing", 2020, arXiv:1911.11947.

[3] The LSST Dark Energy Science Collaboration, "The LSST Dark Energy Science Collaboration (DESC) Science Requirements Document", 2018, arXiv:1809.01669.

[4] E. Saraivanov, H. Rubira, V. Miranda and T. Eifler, "Towards the Two-Loop EFTofLSS in Galaxy Lensing Surveys", 2026, arXiv:2603.13031.

[5] J. DeRose and S. Chen, "The Lensing Counter Narrative: An Effective Description of Small-Scale Clustering in Weak Lensing Power Spectra", 2025, arXiv:2510.18981.

[6] P. McDonald and A. Roy, "Clustering of dark matter tracers: generalizing bias for the coming era of precision LSS", 2009, arXiv:0902.0991.

[7] S. Saito, T. Baldauf, Z. Vlah, U. Seljak, T. Okumura and P. McDonald, "Understanding higher-order nonlocal halo bias at large scales by combining the power spectrum with the bispectrum", 2014, arXiv:1405.1447.

[8] M. Maus, S. Chen, M. White et al., "An analysis of parameter compression and full-modeling techniques with Velocileptors for DESI 2024 and beyond", 2024, arXiv:2404.07312.

[9] T. Lazeyras, C. Wagner, T. Baldauf and F. Schmidt, "Precision measurement of the local bias of dark matter halos", 2016, arXiv:1511.01096.

[10] DESI Collaboration, "DESI 2024 V: Full-shape galaxy clustering from galaxies and quasars", 2024, arXiv:2411.12021.

[11] J. N. Fry, "The Evolution of Bias", 1996, ApJ, 461, L65.

[12] C. Hahn et al., "The DESI Bright Galaxy Survey: Final Target Selection, Design, and Validation", 2023, AJ, 165, 253, arXiv:2208.08512.